

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Emf = 12 V stated [accept $V_{\max}$] (1) Initial current = 2.7 mA (1) $R = \frac{V}{I} = 4\,444\,\Omega$ (1) [4.4 kΩ] [N.B. With incompatible V and I, 2nd and 3rd marks not accessible] $RC \sim 8$ s from 37% $I_{\max}$ value or 63% $V_{\max}$ value or intercept of gradient at the origin with $I = 0$ (or $V = 12$ V) OR taking logs and rearranging OR obtaining Q from total area under current graph around 2.0×10^{-2} C (1) $C = \frac{8.0}{4400} = 1.8$ [± 0.3] mF (1) [No ecf] Unit penalty (–1): all 3 units [V, Ω, F] needed Alternative Emf = 12 V (1) Find $\frac{dV}{dt}$ and I at given t, e.g. at 10 s, 0.411 V s^{-1}, 7.5 mA (1) Hence calculate C from $I = C \frac{dV}{dt} \rightarrow 1.8$ mF ecf (1) $RC \sim 8$ s from 37% $I_{\max}$ value or 63% $V_{\max}$ value or intercept of gradient at $t = 0$ with $I = 0$ (or $V = 12$ V) (1) R from C and $RC \rightarrow 4.4$ kΩ (1) Unit penalty (–1): all 3 units [V, Ω, F] needed		5		5	5	5
			Question 1 total	6	12	0	18	14	7